MACHINE AND BRAKE REPLACEMENT INSTRUCTION FOR MX14 MACHINE IN MACHINE ROOM ELEVATOR

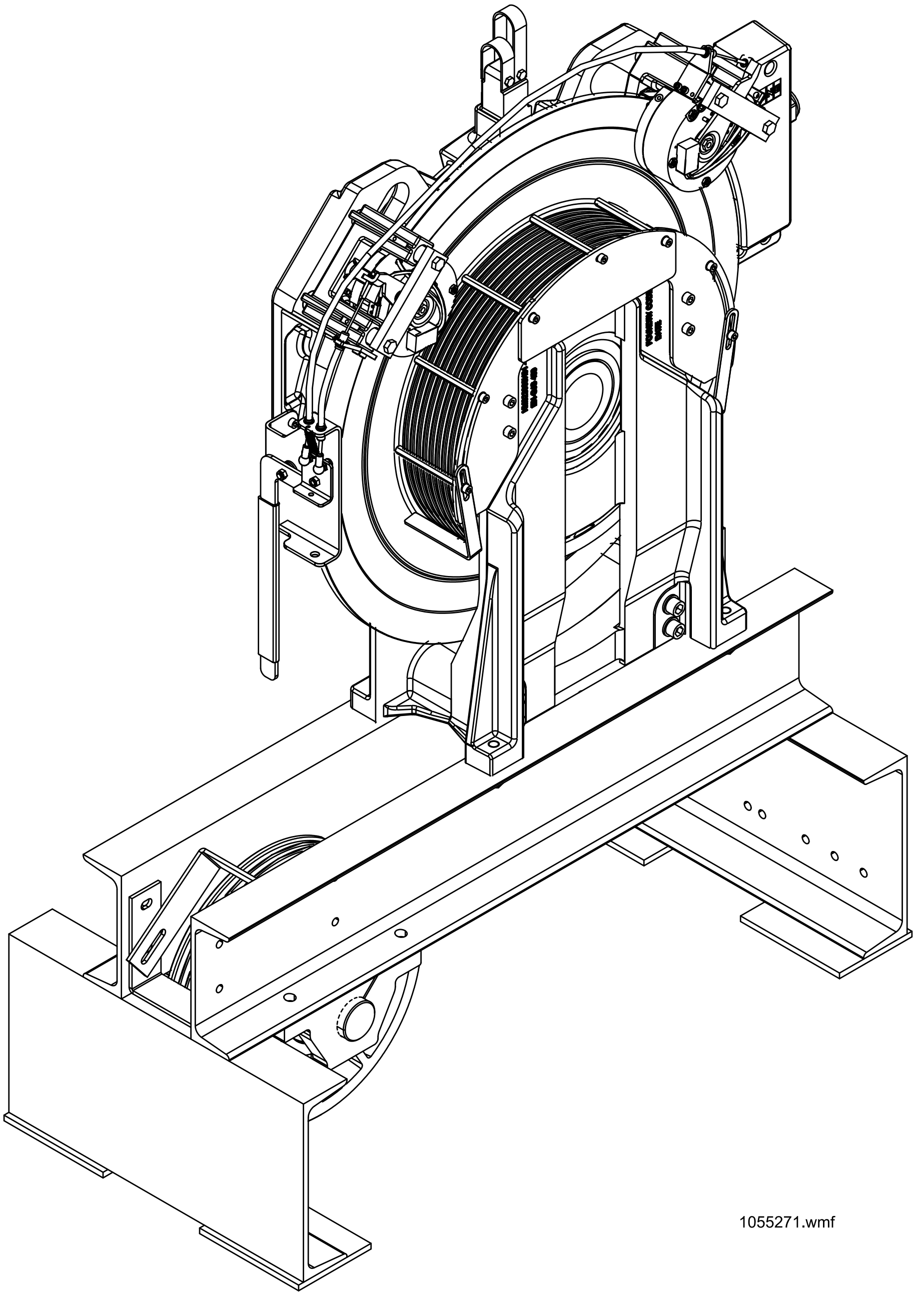

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TABLE OF CONTENTS

1 ABOUT THIS INFORMATION

1.1 Purpose of the information

This information describes the replacement methods and actions for MX14 machine and brake.

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1.2 Audience

This information is intended to be used by people who are familiar with elevator maintenance and have received proper training on methods and safety as specified by KONE.

KONE Destination maintenance and troubleshooting requires special technical skills that must be trained to persons working with the systems. In addition, Front Line (FL) maintenance management must be aware of the content of this information.

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1.3 Related information

- AM-01.03.001 Use of fall arrest systems on elevator construction and modernization sites
- AM-01.03.002 Take 5 - Electrical safety when working on elevators, escalators and autowalks
- AM-04.05.054 Hoisting machine MX14
- AM-09.13.004 Rope tensioning
- AM-11.65.052 KONE N MiniSpace elevators with KDL16S drive, commissioning and safety inspection
- AM-11.65.089 KDM40 and KDM90 drive systems for EN81-20 MiniSpace: Installation, commisioning and safety inspection
- AM-11.65.086 KONE machine room elevators (SOC)with KDL16L / KDL16S drive (EN 81-1 / KCE): commissioning and safety inspection
- AM-11.65.041 KONE 3000S/X/E MiniSpace™ commissioning with KDL16 drive and safety inspection
- AR-04.06.009 Spare parts manual for MX14 machine (KM934300AR01)
- AS-01.03.101 On-site Maintenance Safety Manual
- AS-04.07.003 Replacement instruction for encoder / resolver unit of MX14 machine
- AS-01.01.119 EN 81-20 code and lifts directive 2014/33/EU in KONE elevators
- AS-04.08.092 User instruction for electrical brake opening tool KM872759G01 for MX, GMX and NMX machines
- AS-04.05.039 Rope groove geometry checking
- AS-09.04.002 Rope groove profile checking gauge KM500174G01
- AS-09.05.001 Lubrication of suspension ropes

– AS-09.06.002 Rope faults and criteria for replacement

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1.4 Feedback

Report errors in the information or suggest improvements to *ktd@kone.com*.

1.5 Translations

If you need assistance in getting technical information translated, contact *ktd@kone.com*.

1.6 Approvals and changes

Issue: B

Date: 2018–07–18

Reference CR:

Changes:

– Added Appendix Safety test

– Added chapter Returning Elevator to Service

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– Miska Philippe, Head of Operations Development

Translation approved by:

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2 SAFETY MEASURES

2.1 Safety components

2.1.1 Traceability of safety components

KONE as an economic operator (installer, manufacturer, distributor and importer) keeps a record of suppliers and customers for each installed and replaced safety component. If needed, this enables finding and organizing an immediate replacement of a faulty safety component in any elevator.

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2.1.2 Identification of safety components

KONE safety components can be identified as follows:

– KONE safety components are marked with a CE marking label and orange labels (or similar).

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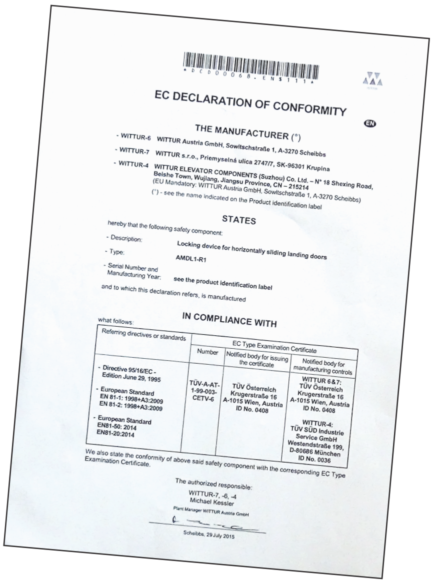
– Safety components have a declaration of conformity document (available in electronic format and sometimes delivered with the safety component).



EC DECLARATION OF CONFORMITY

THE MANUFACTURER (°)

- WITTUR-6 WITTUR Austria GmbH, Sowitschstraße 1, A-3270 Scheibbs
- WITTUR-7 WITTUR s.r.o., Priemyselná ulica 2747/7, SK-96301 Krupina
- WITTUR-4 WITTUR ELEVATOR COMPONENTS (Suzhou) Co. Ltd. – N° 18 Shexing Road, Beishe Town, Wujiang, Jiangsu Province, CN – 215214 (EU Mandatory: WITTUR Austria GmbH, Sowitschstraße 1, A-3270 Scheibbs)

(°) - see the name indicated on the Product identification label

STATES

hereby that the following safety component:

- Description: **Locking device for horizontally sliding landing doors**
- Type: **AMDL1-R1**
- Serial Number and Manufacturing Year: **see the product identification label**

and to which this declaration refers, is manufactured

IN COMPLIANCE WITH

what follows:

Referring directives or standards	EC Type Examination Certificate		
	Number	Notified body for issuing the certificate	Notified body for manufacturing controls
- Directive 95/16/EC - Edition June 29, 1995 - European Standard EN 81-1: 1998+A3:2009 EN 81-2: 1998+A3:2009 - European Standard EN81-50: 2014 EN81-20:2014	TÜV-A-AT-1-99-003-CETV-6	TÜV Österreich Krugerstraße 16 A-1015 Wien, Austria ID No. 0408	WITTUR 6&7: TÜV Österreich Krugerstraße 16 A-1015 Wien, Austria ID No. 0408 WITTUR-4: TÜV SÜD Industrie Service GmbH Westendstraße 199, D-80686 München ID No. 0036

We also state the conformity of above said safety component with the corresponding EC Type Examination Certificate.

The authorized responsible:
WITTUR-7, -6, -4
Michael Kessler
Plant Manager WITTUR Austria GmbH

Scheibbs, 29 July 2015

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2.1.3 Replace safety components

A list of the elevator's safety components is available as a part of the elevator's technical documentation in KONE PDM.

Safety Component List

Safety Component List

ES - MonoSpace Std

KEN	42492015
SAP Sales Order	
Customer Name	
Unit Street	
Unit City	
Country	
Unit Building Name	
Rated load................(kg)	480
Rated speed.............[m/s]	0.63
Elevator: number of floors	3

Type	Description	KM	Serial no	Certificate	Provider	DoC
Brake (Acop and UCMP)	BRAKE, MX05	KM781319G01	HSR031293	2213 CT-04.01.020	Inspecta Tarkastus Oy Inspecta Tarkastus Oy	95/16/EC EN 81-1
Brake (Acop and UCMP)	BRAKE, MX05	KM781319G02	HSR030859	2213 CT-04.01.020	Inspecta Tarkastus Oy Inspecta Tarkastus Oy	95/16/EC EN 81-1
Buffer	SAFETY BUFFER 100X80 ACLA 300501 MODEL D	KM1358184		0044 CT-05.01.200	TÜV	95/16/EC EN 81-1 EN 81-2

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1. Verify that the new spare part has a CE marking.
2. Replace the component.
3. Perform the commissioning tests according to EN 81-1 or EN 81–20 and component specific AS-documents.
4. If the declaration of conformity document is delivered with the spare part, store it in the elevator's logbook folder on-site.
5. Mark the safety component replacement in the elevator's logbook.
6. Backreport the replacement with KONE Field Mobility (KFM) or in a locally agreed way.
7. Demolish the old part beyond repair before disposal.

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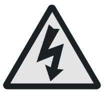
2.2 General safety precautions

Safety precautions	Note
Follow your national elevator safety codes and other safety-related regulations.	In case of conflict between the Code and the present instructions, rely on your national code.
Follow your local hot work procedures.	

Safety precautions	Note
The local safety codes and rules must be obeyed always.	Refer to your local procedures for the type of entrance protection required. **Use barriers and warning signs to inform and protect the building users.**
Follow the safe working methods.	
Follow this instruction. Do not skip any step, otherwise there may be a potentially dangerous situation which you have not considered.	Warning signs highlight possible hazards.  *X000009013*
ENSURE THAT ELECTRICAL EQUIPMENT AND CONDUCTORS ARE SAFELY DE-ENERGISED BEFORE WORKING ON THEM. A locking off system for main electric supply isolator or other system (for example, fuse removal, locking and tagging system when applicable) must be agreed with person responsible for the building electrification.	Do not connect or disconnect any connectors when the power is ON.  *X000009007*
Personal protection equipment must be available and used as required.	Refer to the list of Personal safety items.
IF RISK OF INJURY FROM A FALL **Adequate fall prevention system must be in place.**	 *X000009019* For using fall prevention systems, refer to ***AM-01.03.001***.
If the safety harness is used incorrectly, you may still sustain severe injury or death during a fall or uncontrolled movement of the elevator. If you have attached the harness to a guide rail bracket or other fixed point in the elevator shaft, you must ensure that there can be no unwanted movement of the car or counterweight.	
Handle and dispose of waste materials in accordance with the regulations applicable to your country/state.	
Ensure that your work does not cause a hazard to others.	In particular keep access ways and fire exits clear. Refer to AS-01.01.004 Take Care End user safety in elevator maintenance[1]
Keep the entrance and emergency exits clear. Prevent unintentional access to working area with extra fences.	

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2.3 Personal safety

Safety gloves, overall, safety shoes with ankle protection, safety helmet, safety goggles, dust mask, hearing protection and safety harness are provided for your personal protection. USE THEM AS REQUIRED.

1) AS-01.01.004 has been retired and replaced by AS-01.03.101 Elevator and building door maintenance - safety manual

Safety item	Figure
Fall preventation equipment. Refer to **AM-01.03.001**.	
Dust mask, suitable for working with mineral wool insulation in landing doors	
First aid kit	
Safety goggles	
Safety gloves	
Rubber gloves for cleaning rails	
Hearing protection	
Hard hats	
Work clothes / overalls	
Safety shoes with ankle protection	

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2.4 Maintenance method safety

Note the following safety items when working with MX14 machines:

- Site survey must be done with supervisor before starting the first maintenance work.
- Follow all the safety rules and instructions concerning general working with elevators.
- Refer to *AS-01.03.101* for safe procedures to prevent hazards related to moving between the landing and the elevator shaft, working on the elevator car roof, driving on inspection and working in the elevator shaft pit.
- Refer to AM-01.03.002 Take 5 - Electrical safety when working on elevators, escalators and autowalks. The Take 5 safety initiative is designed for installation, servicing, maintenance and modernization work done on the elevators.
- The car must always be held by two different supports, if the ropes are slack or removed. A secured safety gear and a parking chain are to be used (one or two chains depending on weight of car and suitability of guide rail brackets). The parking chains are routed over the nearest guide rail support beam to the car sling.
- If there is any risk to the building users caused by unguarded landing entrances, use barriers and other safety measures to protect the area around the topmost landing door.
- Use gloves when handling the ropes in order to avoid injuries from rope splinters.
- MX14 transportation to the machine room must be planned beforehand case by case.
- Ensure that the MX machine package is never placed on an uneven or inclined surface. If tilted the machine may fall over.
- Loading capacities must be checked before moving the MX14 machine (e.g. lifting points, floors and staircase).

- Take special care when lifting heavy components like machine. When loosening any fixings, ensure that the remaining supports can carry all the loads bearing on them.
- All hoisting of MX14 machine must be done from the two eye bolts at the side of the machine. The eye bolt between the brakes must not be used for lifting the machine.
- An additional screen may be required if an adjacent machine is running in common machine room.
- The cover of the drive module must be kept closed always when not working with the module, even if the main power has been switched OFF.
- The control cabinet door and the cover of the drive module must always kept closed. Even, when the main power is OFF, or when not working with the panel.
- Prevent the reconnection of auxiliary emergency power supplies etc.
- Be careful when opening screws and nuts to avoid dropping metal items into the pulley housings. Although the housings are protected with covers and brushes, it is possible that small particles catch in the pulley grooves.

WARNING: The MX motor will act as a generator, when the motor is rotated for example manually.

Ensure that the motor wires are insulated. Ensure cover is fitted to motor terminal box and that you cannot come into contact with exposed conductors.

WARNING: Inverter drives usually remain energized for about 5 minutes after the power has been disconnected.

DO NOT work on the drive, hoisting motor or braking resistors until you have verified that this energy has been discharged.

Test equipment must be set to the 1000 VDC range. The test equipment must be checked before and after the test to ensure that it functions correctly.

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2.5 Magnetic field

The machines are surrounded by a strong magnetic field when the motor body is opened and the rotor uncovered. Because of the magnetic field, electronic devices may malfunction or cease to operate in the vicinity of the MX machines.

WARNING: STRONG magnetic attraction force!

If you use a pacemaker, a hearing aid or other kind of sensitive electronic device keep away from the working area.

Be careful when working or handling metallic objects in vicinity of open rotor

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2.6 Working with MX machines

Because of the construction of the MX-machines note the following:

– MX-type machine is equipped with a permanent magnet AC-motor.
– Because of this feature magnetizable dust from sources such as welding, grinding, gravel or similar must be kept outside the machine.
– When there is a danger of magnetizable dust getting inside the machine, use always a reinforced adhesive tape to cover the air gap between the rotor and the machine frame.
– Do not drill any holes to the motor body (drilling is allowed only to the bars on the body sides).
– Ensure the tools you use are suitable for lifting and moving big and heavy parts. The tools must be tested and inspected according to local rules.
– Check the loading capacity of the machine room floor before moving the machine.

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2.7 Environmental issues

Handle hazardous waste and material according to local regulations and KONE Environmental Strategy.

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3 MX14 MACHINE AND BRAKES

The MX14 machine is a gearless hoisting machine for elevators with or without machine room. The machine weighs approximately 515 kg.

The machine has two independent disc brakes which do not need to be adjusted after the brake replacement. The brake units weigh approximately 20 kg each.

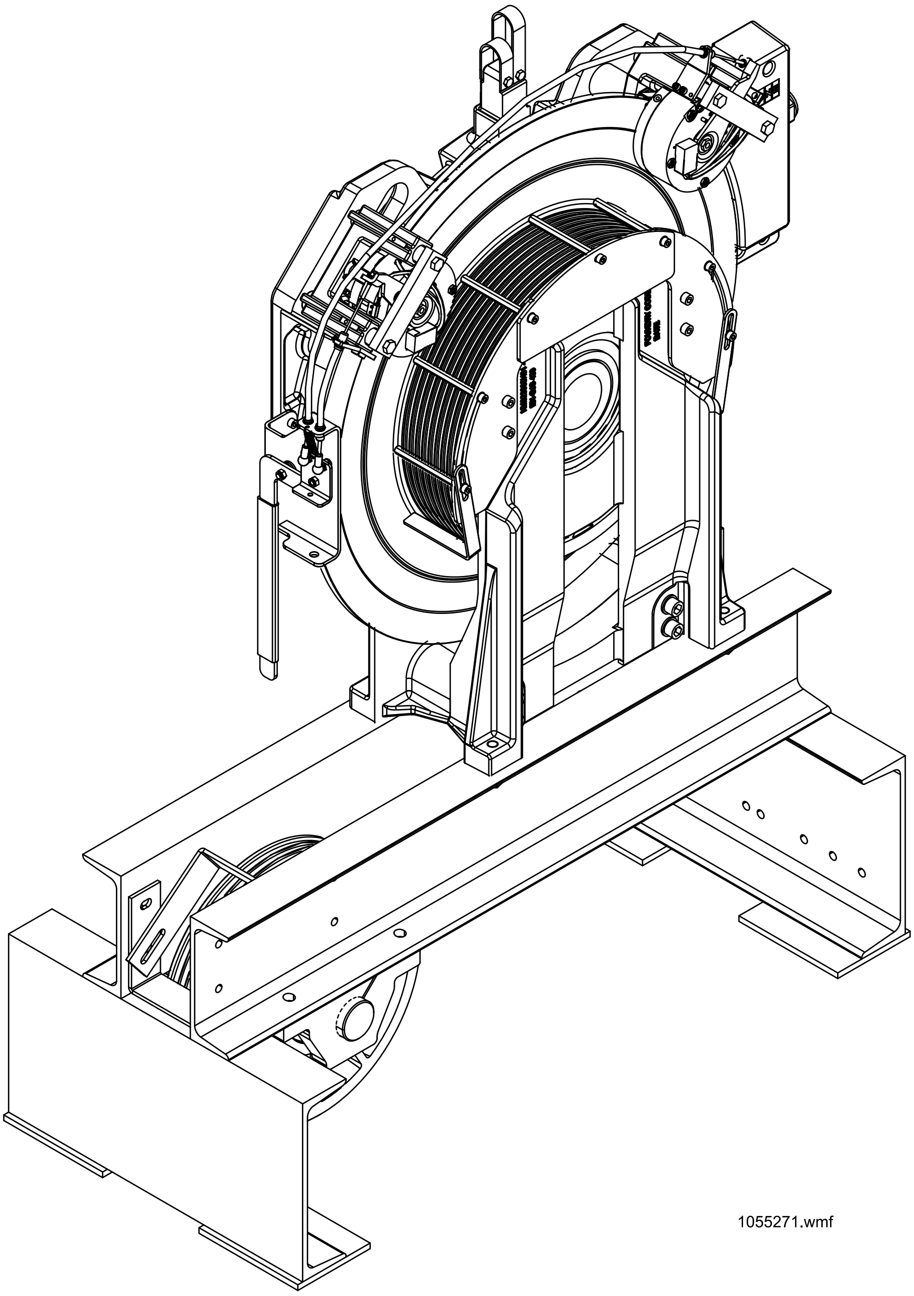

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Figure 1: MX14 machine in machine room

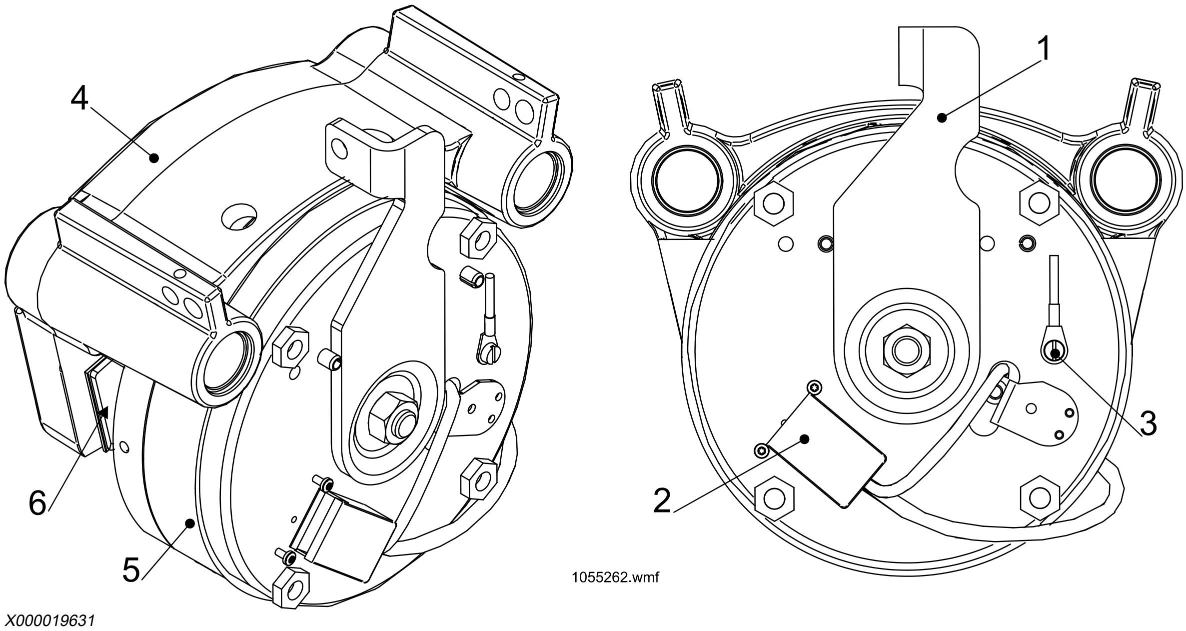

Figure 2: Machine brakes

1. Brake lever
2. Brake electrification plug connector
3. Earthing wire
4. Body
5. Magnet
6. Brake shoe

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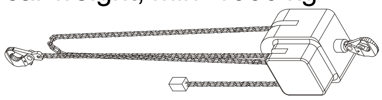
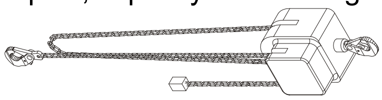
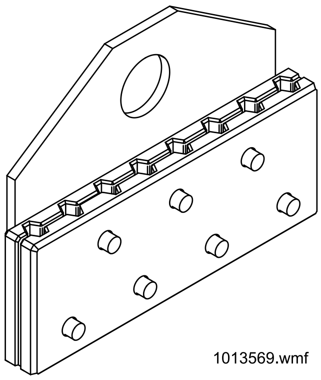
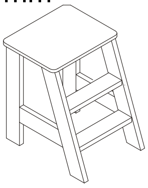
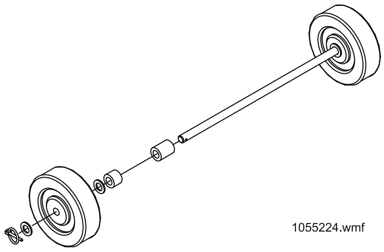
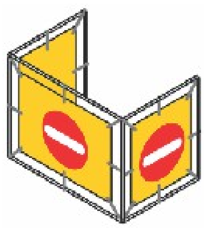
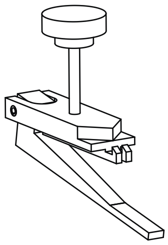
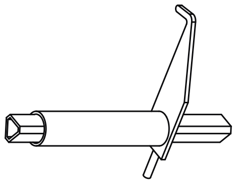
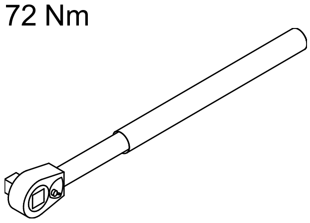
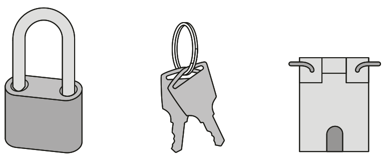
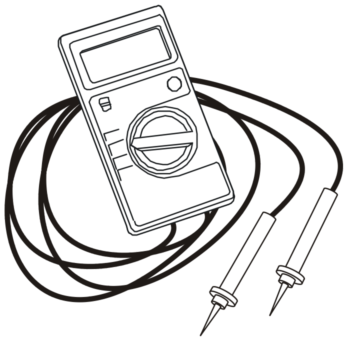
4 TOOLS

Part Number	Tool	Illustration / Note
KM50009502R01	Standard maintenance tool set	 X000080701
Locally provided	Chain hoist or equivalent for lifting car	1 pcs., capacity according to car weight, min. 1000 kg  X000105467
Locally provided	Chain hoist or equivalent for suspending machine	2 pcs., capacity min. 500 kg  X000105467
KM266826	Adjustable hoisting chain	Capacity according to car weight, L = 1,5 m  X0000108509
KM266827	Hoisting chain	Min. capacity 1500 kg, L = 1,5 m X0000108467
Locally provided KM266828	Parking chain	2 pcs., min. capacity according to car weight  X000069637
KM717384G01, d8 mm ropes KM762959G01, d10 mm ropes	Rope clamp	 X000069633

Part Number	Tool	Illustration / Note
KM265173	Work stool	Height approximately 700 mm  X000105472
KM1355865 KM50020169	Pit prop Locally delivered, e.g.: Peri adjustable multiprop 250 Peri multiprop MP 50 (extension)	2 pcs., rated load according to site survey  X0000115608
Locally provided	Test weights	Totaling 125% of rated load.
Locally provided	Bolt Washer Nut	M6x125 full thread DIN9021, D6.4 mm, 2 pcs. For brake test
KM789545G02	Machine transporting wheels (3 pcs.) for use with MX14 transportation frame	 X000010943
Locally provided.	Safety fences	 X0000085588

Part Number	**Tool**	**Illustration / Note**
KM871952G01	Door blocking tool	 *X000013258*
KM748001G01, L=200mm KM760860G01, L=700mm KM748001G04, L=1100mm	Emergency opening key	 *X000012478*
Locally provided	Cleaning cloth	
KM500174G01	Gauge, for measuring pulley groove profile	
Locally provided	Vacuum cleaner	
Locally provided	Torque wrench Socket key set	1/4”, for rotor bolts, torque set to 72 Nm  *X000069636*
Locally provided	Maintenance signs	 *X0000070609*
KM872759G0	Electrical brake release device	
KM854281	Locking set for main switch	 *X000079905*

Part Number	Tool	Illustration / Note
KM964698	Cut-resistant protective gloves	
Earth resistance meter	Metrel MI 3121 SMARTEC insulation and continuity tester, or equivalent, EN 61557 compliant device	 X000015133

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5 PREVENTIVE MAINTENANCE

5.1 Safety measures

WARNING: Checks that involve risks related to stored energy must be conducted with the main power switched off, locked & tagged and de-energized.

WARNING: Pay special attention to switching off power of correct machinery in group elevators.

NOTE: Refer to *AS-01.03.101* for safe procedures.

– Put Under maintenance signs to all landings.

– Inhibit landing calls and door opening.

– Switch OFF the mains power supply

– Lock and tag the mains switch

NOTE: Verify that the elevator car is empty before beginning of tasks.

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5.2 Maintenance guideline for machines

1. Operate the machine stop button (if applicable).

   If the button is inoperative, but is applicable and required, it must be replaced or repaired.

   If problems exist contact the supervisor to agree on further actions.

2. Check the motor supply cables visually.

   Fix all the motor supply cables properly.

3. Check and connect the connector of the encoder / resolver cable properly.

4. Surface of the brake disc must be clean of grease and rust.

5. Tighten and secure the fixing bolts if necessary. Replace missing lock pins.

   All fixing bolts to the machine bed plate should be tight and secured (corners of the lock plate twisted). The lock pins must be in place.

6. Check the sheave. If problems exist contact the supervisor to agree on further actions.

   Traction sheave must be vertically aligned, maximum allowed deviation between the traction sheave and ropes is 2º.

7. Check and clean the machine and prevent small particles from getting into the machine.

   The machine should be clean outside.
8. Clean with cloth if necessary.

   There should not be excess oil in the ropes because this might drift to brake disc.

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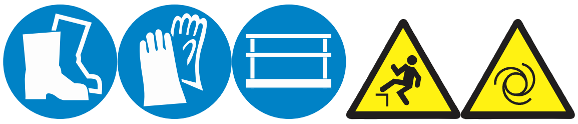
5.3 Maintenance guideline for brakes

The brakes must be inspected once a year or according to local codes.



1. Perform one-sided braking test before and after each air gap measurement.
2. Check the brake air gap. It should be within 0.2 mm - 0.55 mm.
   1. Connect the electrical brake opening tool.

      NOTE: Open one brake at a time.
   2. Move the brake until the entire air gap is on one side of the brake disc drum (the other brake pad touches the brake disc).
   3. Measure the air gap with 0.2 mm and 0.6 mm feeler gauges. The 0.2 mm feeler gauge should fit, 0.6 mm should not (exact limit is 0.55 mm). If the brake fails this test it should be replaced (the adjustment of the air gap is not possible).
   4. Reconnect the brake cable.
   5. Repeat the test with the other brake.

      NOTE: Do not open the brake from the center nut or from the manual brake release lever!
3. Check visually that the brake release cables are ok and that there are no sharp turns or twists.

   Brake release cables should be free of twists or sharp turns. The cable sleeve should also be intact.
4. Check and test brake release operation.

   The brakes must fully open when the brake release lever is pulled.

   WARNING: Close the brakes every 0.5 — 1.0 second to avoid the elevator accelerate to over speed.

   ---
5. Check the brake lever rests on the limiter pin.

6. Move the brake lever on the brake by hand. It should move easily at least 5 mm.

   The brake lever must have at least 5 mm of free movement when the brake is normally closed.

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Related information

– AS-04.08.092 User Instruction for MX electrical brake opening tool
– One-sided electrical brake test (with 0% load) (44)

5.4 Maintenance guidelines for traction sheave and ropes

1. Check the rope and nip guard fixings by shaking them manually.
2. Put the guards back in correct position if necessary. If they are not available, arrange them for the next visit.

   Rope guards must be in position to stop the ropes from jumping off, nip guards should be in correct position.
3. Check that the traction sheave bearing does not make any abnormal noise.
4. Ensure that the ropes sit on the same depth on the rope grooves, maximum deviation is 0.2 mm.
5. Ensure that there is at least 2 mm gap between the rope and the bottom of the undercut. Visible rope marks must not exist on the bottom of the grooves.
6. Check the ropes according to AS-09.06.002.

   If problems exist contact the supervisor to agree on further actions.

   Lubricate ropes if needed. The ropes must not be dry nor show any signs of rust.
7. Check visually. If necessary, investigate the root cause and report to the supervisor.

   No metallic particles should come from the ropes or traction sheave. There should be no visual signs of excessive wear.

   If metallic particles are present, automatic rope lubricators must be installed.

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Related information

– AS-04.05.039 Rope groove geometry checking
– AS-09.05.001 Lubricating the suspension ropes
– AS-09.06.002 Rope faults and criteria for replacement

6 REPLACEMENT OF MX14 MACHINE

The MX14 machine with fixings weighs approximately 515 kg. It is important to use correct tools and correct suspension points for lifting and moving the machine.

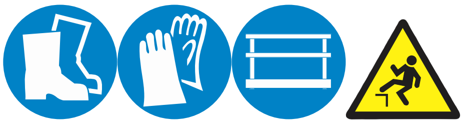
Replacement work requires two maintenance technicians. Use the elevator to transport the new machine. If the elevator is not available, the transportation has to be planned in advance case by case.

Related information

– AM-01.03.009 General instructions for material handling, using transportation and lifting equipment

6.1 Safety measures

NOTE: Refer to AS-01.03.101 for safe procedures.



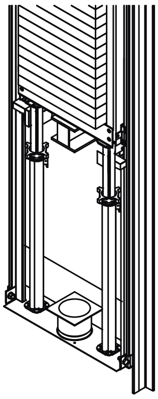
The following safety measures must be carried out before the actual work:

1. Drive the car to the topmost floor and take the elevator out of service.

   NOTE: Install the safety fences around the work area to prevent unauthorized access.

2. Switch on the RDF.
3. Drive the car to suitable position with access to the car roof and topmost landing level.
4. Turn the elevator to inspection.
5. Switch off the RDF.
6. Enter the car roof with chain hoist.
7. Drive the car to suitable position with access to the shaft ceiling.
8. Attach the chain hoist.
9. Drive the car to the topmost landing with access to the car roof and topmost landing level.
10. Go to the landing.
11. Switch off the main switch.
12. Lock and tag the main switch.
13. Close the machine room.
14. Go to the pit.
15. Fit and secure the pit props under the counterweight.

16. Ensure the pit props are vertically aligned and centre to the frame.



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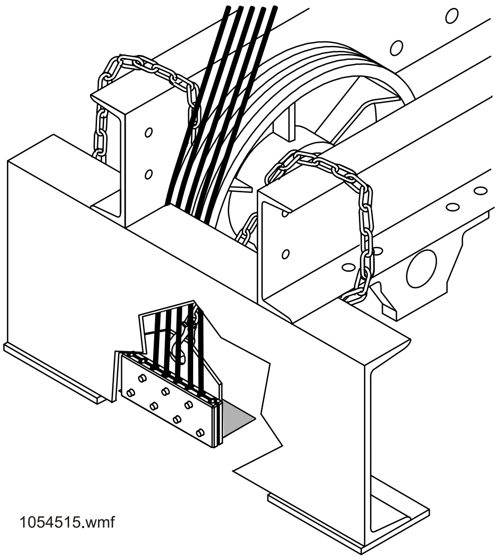
17. Attach a rope clamp to the counterweight side ropes.
18. Secure the rope clamp to the bed plate by using adjustable hoisting chains.



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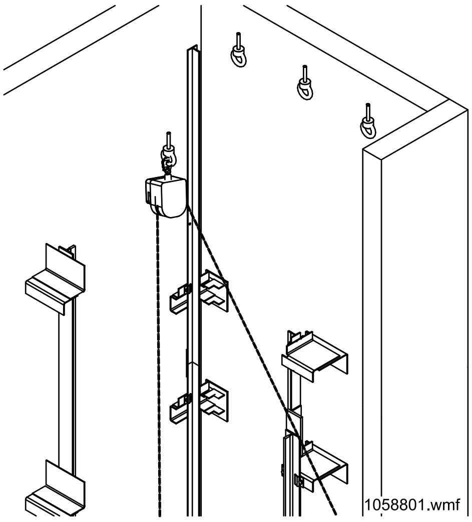
19. Attach the chain hoist to the car sling.

20. Lift the car approximately 200 mm.

CAUTION: The car must be lifted from the landing



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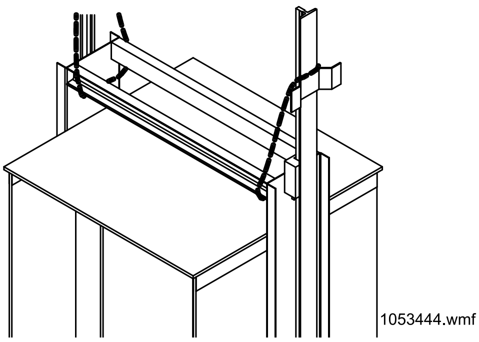
21. Lower the car onto the safety gear.
22. Secure the car by using the parking chain(s), depending on car weight, over the nearest support beam of the guide rail bracket to the car sling.

    NOTE: Adjust the parking chain as tight as possible. Ensure that the ropes are loose.



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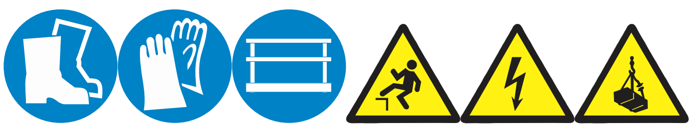
Related information

– AS-01.03.101 On-Site Maintenance Safety Manual

6.2 Remove old machine

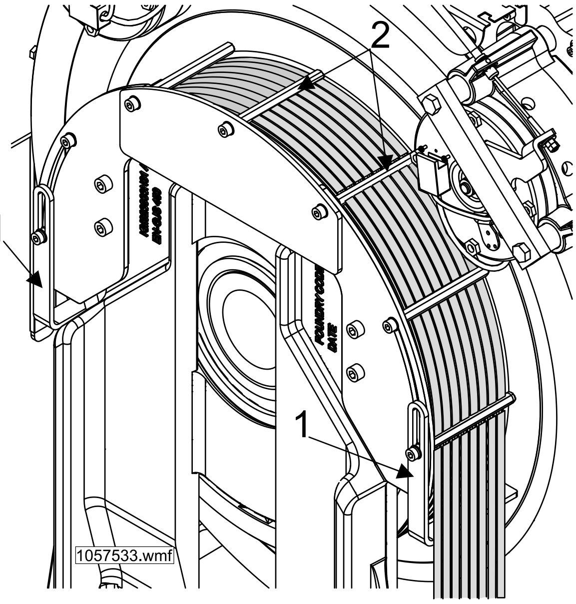


1. Disconnect wiring from the machine.

WARNING: Ensure that the power has been switched off and de-energization is verified.

---

2. Remove the nip guards (1) and rope guard (2).



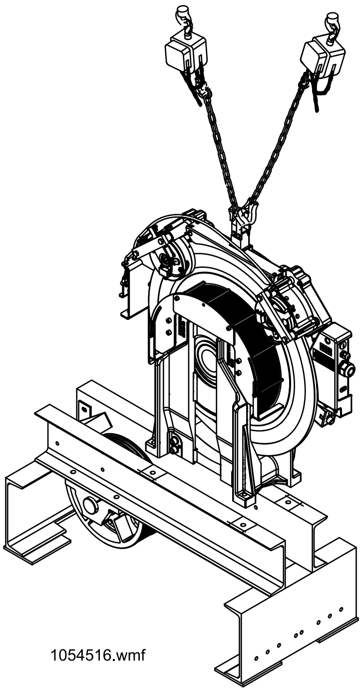
X000019779

3. Mark the position of the machine to the bed plate.
4. Mark the position of the bed plate to the machine room floor.
5. Take the ropes off the traction sheave.
6. Attach the chain hoists to the machine room ceiling.
7. Attach the chains to the centre eye bolt of the machine.
8. Tighten the chains as tight as possible.

WARNING: Do not go under the suspended load.

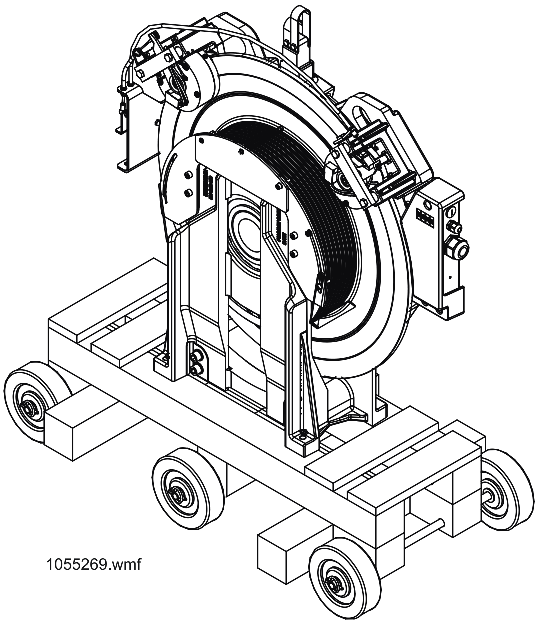
9. Remove the machine fixing bolts.

10. Lift the machine off the bed plate.



X000019780

11. Place the transportation frame beside the machine.
12. Lift the machine onto the transportation frame, and secure the machine to the frame with coach screws.
13. Remove the hoisting chains from the machine.



X000019781

14. Transport out the machine.

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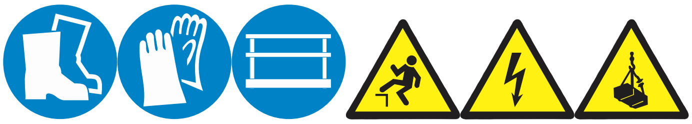
Related information

– AM-01.03.009 General instructions for material handling, using transportation and lifting equipment
– De-energize KDL16R drive (56)

6.3 Install new machine



CAUTION: Beware not to tilt the MX machine because it may fall over.

1. Transport the new machine to the machine room alongside the bed plate.

   NOTE: Ensure that resolver/encoder unit is not damaged.

2. Attach the hoisting chains to the machine lifting eyes.
3. Secure the loose ropes alongside the bed plate.
4. Lift the machine onto the bed plate.

   NOTE: Keep the hoisting chains tight until the machine is fixed to the bed plate.

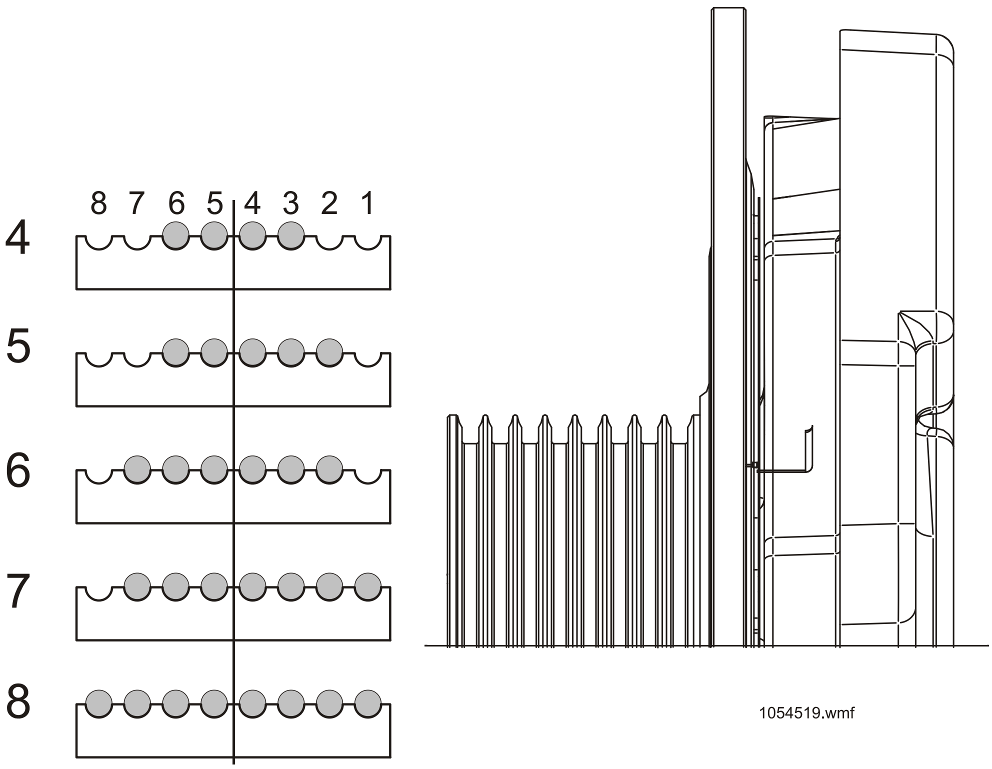
5. Align the machine to the previously marked position.
6. Install the machine to the bed plate with fixing bolts.
7. Reconnect all the cables to the machine.

   WARNING: Power must be switched off and de-energized.

8. Do the commissioning before reinstalling the ropes.

9. Lift the ropes back to the traction sheave, and remove the rope clamp.

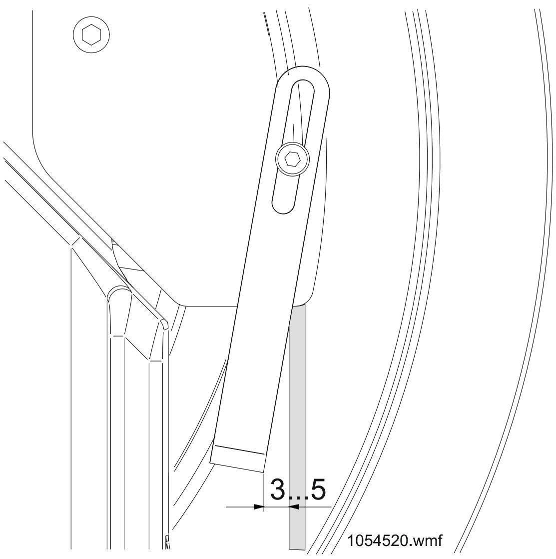
   NOTE: Machine room version machines use 4 – 8 ropes. Ropes have to be placed in the middle of the traction sheave. When the number of ropes is odd (5 or 7) one extra groove is occupied towards interior of the machine and an empty one is left towards the exterior.



X000018588

10. Reinstall the rope guards.

    Adjust the gap between the nip guards and ropes to 3mm.



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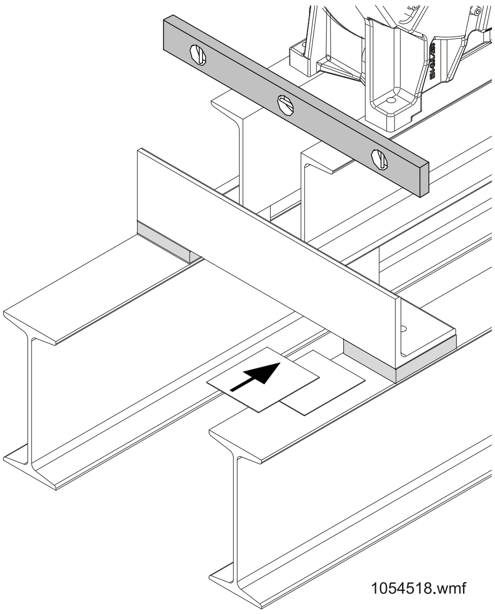
11. Lift the car out of the safety gear.
12. Remove the parking chain(s).
13. Remove the pit props.
14. Check alignment of the bed plate.

    If necessary, adjust by adding or removing shim plates.



X000019782

15. Do the commissioning after roping.

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Related information

– AM-11.65.052 KONE N MiniSpace elevators with KDL16S drive, commissioning and safety inspection
– AM-11.65.089 KDM40 and KDM90 drive systems for EN81-20 MiniSpace: Installation, commisioning and safety inspection
– AM-11.65.086 KONE machine room elevators (SOC) with KDL16L / KDL16S drive (EN 81-1 / KCE): commissioning and safety inspection
– AM-11.65.041 KONE 3000S/X/E MiniSpace™ commissioning with KDL16 drive and safety inspection
– AM-01.03.009 General instructions for material handling, using transportation and lifting equipment

6.4 Finalize the machine replacement

1. After the electrification has been completed and the power is turned on, check the operation of the brakes.
2. Move the elevator by using RDF drive in up and down directions. Push the stop button, the car must not move.

   NOTE: If the car moves, contact your supervisor for the adequate expertise and guidance. The brakes may need to be changed.

3. Perform one-sided braking test.
4. Check the rope tension.
5. Perform the safety tests.

   NOTE: According to EN 81-20 standard, the elevator must pass the safety tests before it is returned to service.

   WARNING: If any of the safety tests fail, the elevator must not be put into normal use before it has been repaired and passed the tests.

X0000188247 A.6

Related information

– Preventive Maintenance
– AM-09.13.004 Rope Tensioning
– Safety tests (44)
– One-sided electrical brake test (with 0% load) (44)

6.5 Return elevator to service

1. Record the maintenance work to the maintenance log book and fill the test report.

   NOTE: Update the maintenance log book with the following details regarding component change:

   – Serial number, if available
   – Component KM number

   NOTE: Back report the replacement with KFM or in a locally agreed way.

2. Collect the reading of the start counter and record the amount to logbook.
3. Dispose of the old machine and clean the site.

   NOTE: Disposal of the old machine shall be performed according to replacement plan.

4. Reset the faults from the controller.
5. Perform a test run.
6. Enable landing calls and door opening.
7. Remove safety fences and "Under maintenance" signs from landings.

X0000189325 A.4

Related information

– AS-01.01.190 Environmental excellence - Maintenance

X0000188158 A.5

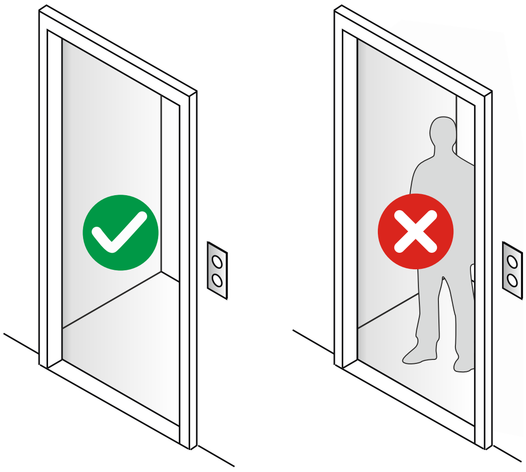
7 REPLACEMENT OF MX14 BRAKE UNIT

7.1 Safety measures

NOTE: Refer to AS-01.03.101 for safe procedures.

The following safety items must be carried out before the actual work.

1. Transport the new brakes to the machine room.
2. Ensure that the elevator car is empty.



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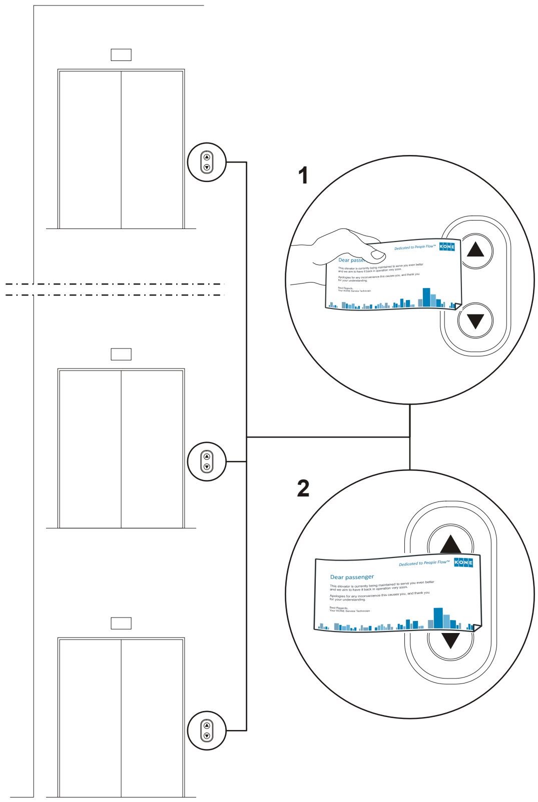
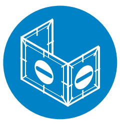
3. Put under maintenance stickers at all landings.



X000013324

Install the safety fences around the work area to prevent unauthorized access.



4. Lower the counterweight to the buffer.
5. Turn off the main switch.

   Lock and tag the main switch.

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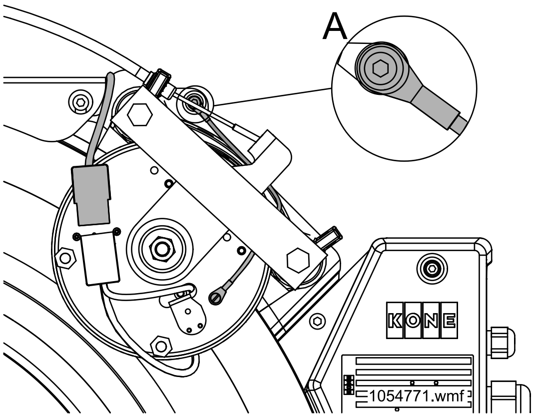
7.2 Remove brake unit

WARNING: Do not remove both brakes simultaneously.

---

1. Disconnect the brake cable.

2. Disconnect the brake earthing wire (A).



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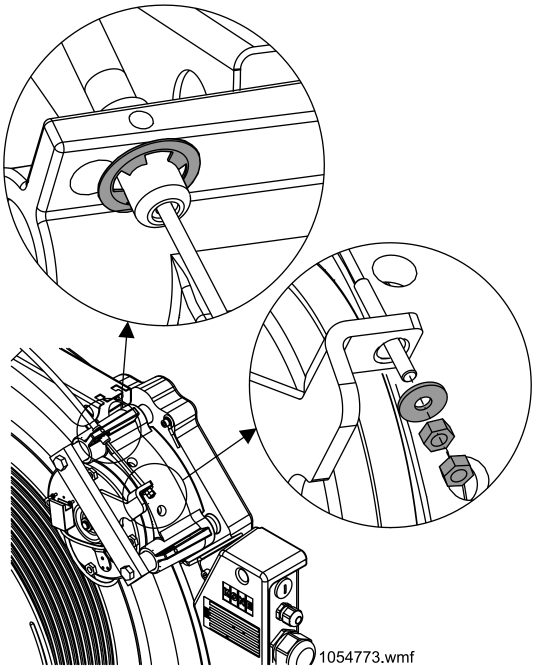
3. Disconnect the brake release cable from the brake.



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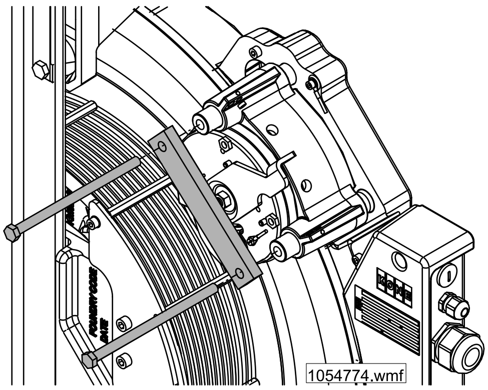
4. Remove the brake fixing bolts and the support plate.



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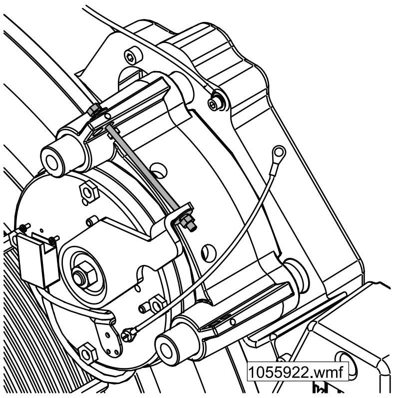
5. Open the brake with M6x125 mm bolt, nut and washers (2 pcs.).



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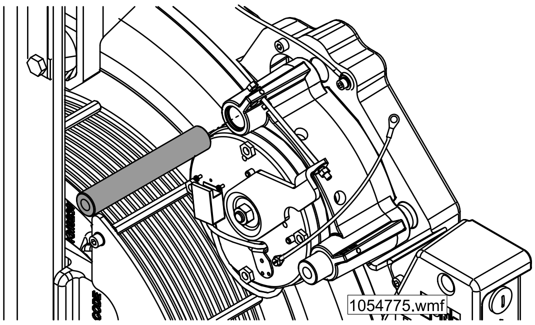
6. Remove the upper pin while holding the brake firmly.



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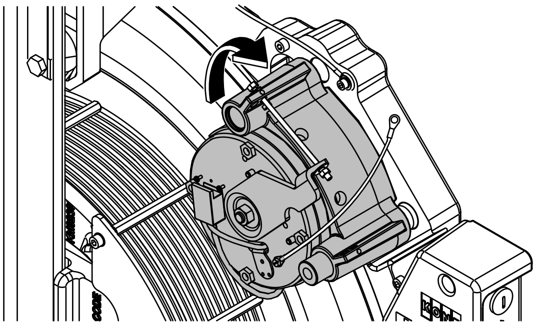
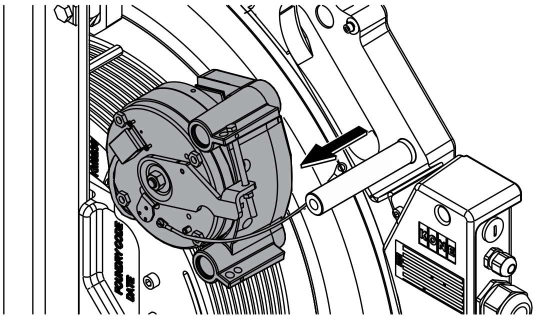
NOTE: The upper end of the brake becomes loose when the pin is removed.

7. Turn the brake upwards and pull it out.



X0000111523



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NOTE: If the brake unit does not move freely, clean the brake pin. Check for possible damage and use fine grade sand paper or emery cloth to remove the scratches.

Ensure that metal particles cannot get into the machine.

8. Place the brake on a solid surface.

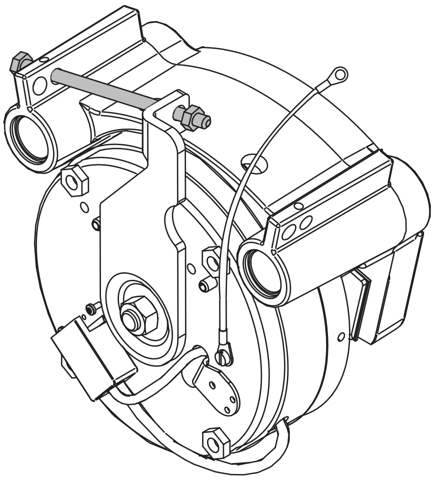
NOTE: Ensure that nothing gets inside the brake when the brake closes.

X0000114951 A.10

7.3 Install and adjust brake unit

NOTE: The maintenance person must verify CE marking in the safety component, store the Declaration of Conformity document in the logbook folder in the site and enter the safety component changes in the elevator logbook.

1. Open the new brake with M6x125 mm bolt, nut and washers (2 pcs.).



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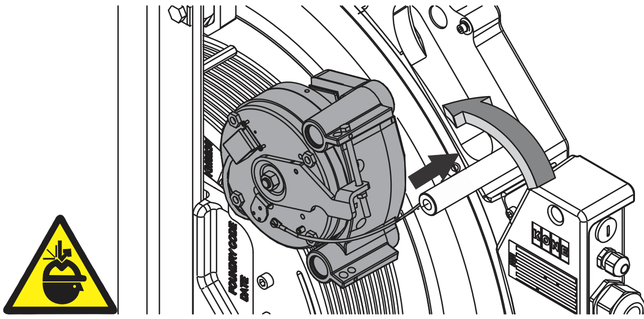
2. Install the brake to the lower brake pin and slide it inwards.



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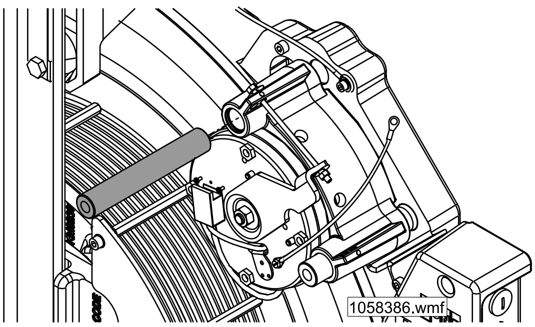
NOTE: Check that the new brake unit is correct type and does not have transportation damage.

3. Turn the brake downwards until the brake disc is between the brake pads.

NOTE: Do not the damage the brake pads against the brake disc.

4. Install the upper brake pin.



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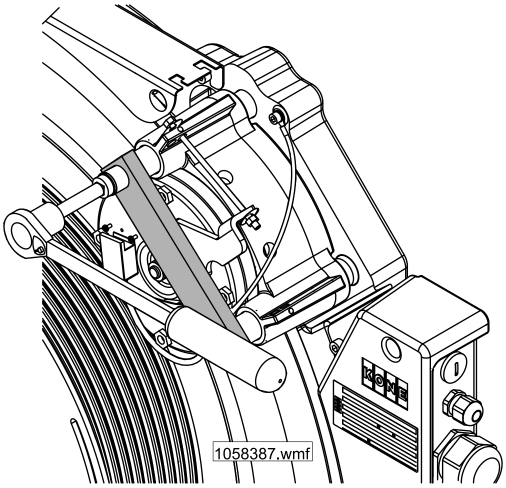
5. Install the brake support plate and the brake bolts.



X000019643

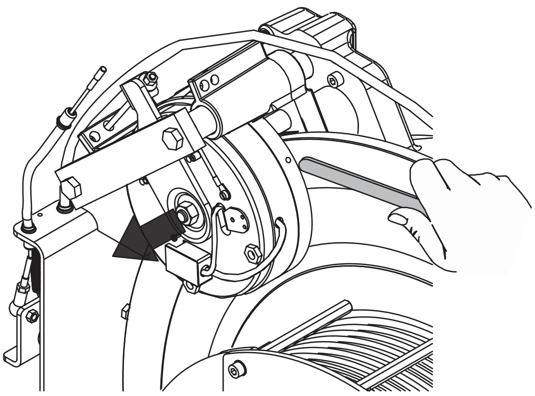
NOTE: Tighten the brake bolts. Torque 72 Nm.

6. Move the brake body by hand.

   The brake must move slightly on the brake pins only when the brake is open (this is the clearance from the air gap).

   If the brake is open and it does not move freely, check the brake fixing.

7. Move the brake by hand until the entire air gap is on one side of the brake disc and other brake pad touches the brake disc.



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Measure the air gap with 0.2 mm and 0.6 mm feeler gauges (for example, 0.5 + 0.1 mm). The 0.2 mm feeler gauge must fit, 0.6 mm must not fit (exact limit is 0.55 mm).

NOTE: If the brake does not pass this test, the brake must be replaced.

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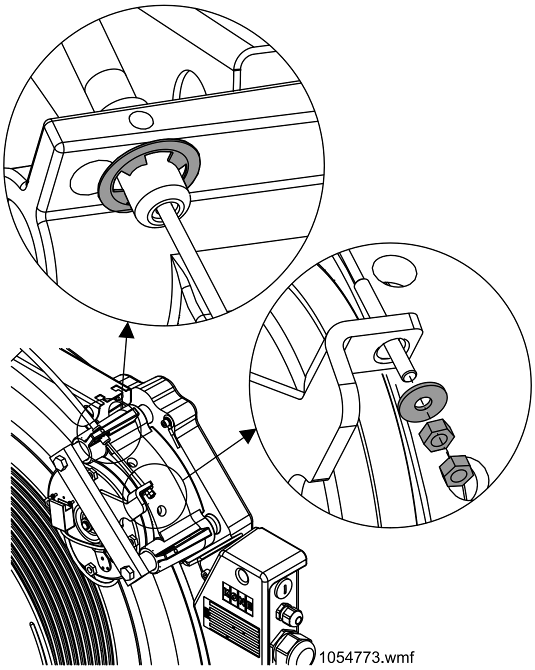
(B) 2018-07-18

8. Remove the bolt, nut and washers (2 pcs.). Connect and adjust the brake release cable.



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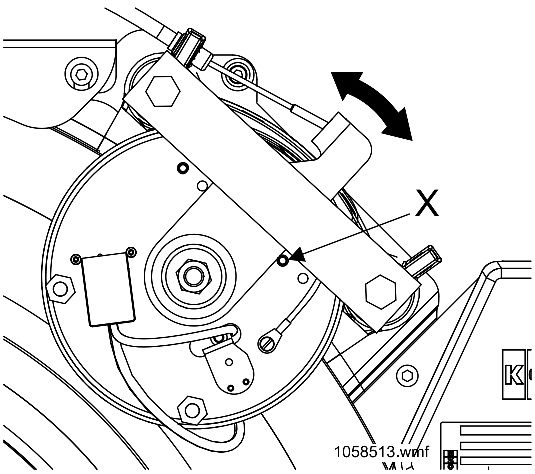
9. Check that the brake lever rests on the limiter pin (X).



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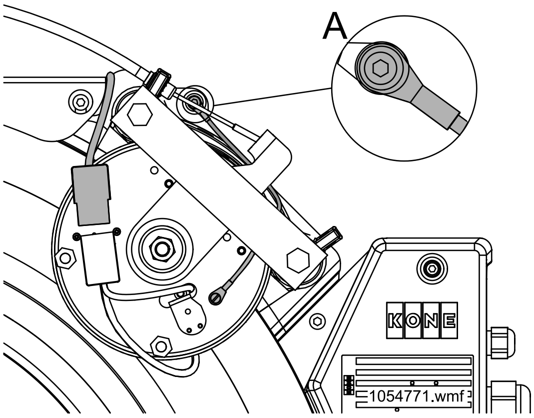
Move the brake lever by hand. The lever must move easily for at least 5 mm.

NOTE: This check ensures that brake spring force goes to the friction pads, not to manual opening mechanism.

10. Connect the brake earthing wire (A).

    NOTE: Verify earth continuity. See section Measure protective earth continuity.



X000019634

11. Connect the brake cable.

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Related information

– AM-11.65.052 KONE N MiniSpace elevators with KDL16S drive, commissioning and safety inspection
– AM-11.65.089 KDM40 and KDM90 drive systems for EN81-20 MiniSpace: Installation, commisioning and safety inspection
– AM-11.65.086 KONE machine room elevators (SOC) with KDL16L / KDL16S drive (EN 81-1 / KCE): commissioning and safety inspection
– AM-11.65.041 KONE 3000S/X/E MiniSpace™ commissioning with KDL16 drive and safety inspection

7.4 Test the brake after replacement

NOTE: Refer to AS-01.03.101 for safe procedures.

1. Turn on the main switch.
2. Test the operation of the manual brake releasing device.

WARNING: Close the brakes every 0.5 — 1.0 second to avoid the elevator accelerate to over speed.

For more information, refer to *Preventive Maintenance*.

NOTE: Use the electrical brake release device (RBO), if applicable. For more information, refer to AS-10.60.100.

3. Perform the following safety tests.
   – One-sided electrical brake test (with 0% load)
   – Single brake test with rated speed and 100% load (EN 81-20 6.3.1 b)
   – Brake test with 125% load (EN 81-20 6.3.1 a)

   For more information, See *Safety tests*.

   NOTE: According to EN 81-20 standard, the elevator must pass the safety tests before it is returned to service.

   WARNING: If any of the safety tests fails, the elevator must not be put into normal use before it is repaired and passes all the tests.

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7.5 Finalize replacement

1. Mark the maintenance work to the elevator log book and fill the test report.

   NOTE: Update the elevator logbook with the following details regarding component change:
   – S/N if available
   – Component KM number

   NOTE: Back-report the replacement with KFM or in a locally agreed way

2. Clean the site.

   NOTE: Dispose the old components according to the instructions in *AS-01.01.190*.

3. Return the elevator to service and perform a test run.
   – Enable the landing calls and door opening.
   – Remove safety fences and "Under maintenance" stickers from the landings.

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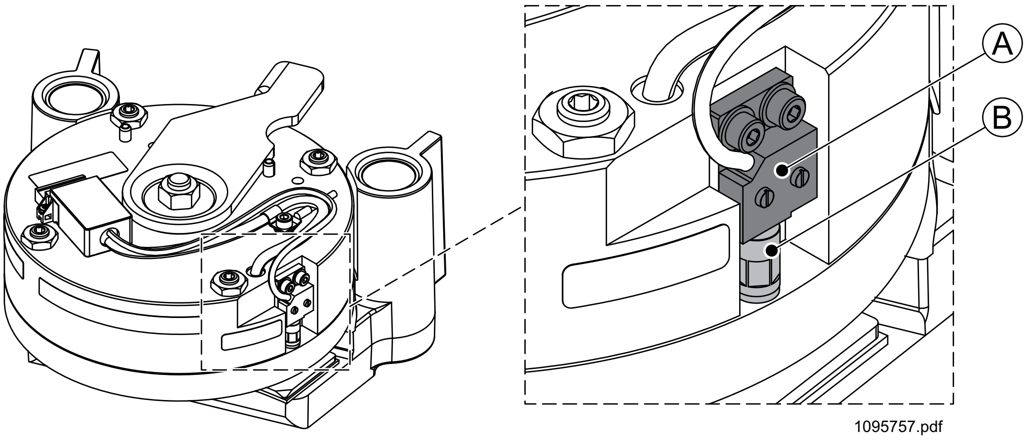
APPENDIX A. OPTIONAL PROXIMITY SWITCH FOR MONITORING THE BRAKE OPERATION

The non-contact proximity switch (A) (optional) monitors the brake operation. It detects the presence or absence of plunger (B) in the brake and signals the state of the brake. The proximity switch has normally open configuration (when the brake is open, the output signal is in on state).

NOTE: The proximity switch is installed, adjusted and tested in the manufacturing. It should not require any checking or adjusting at site. If required, for more information on adjusting or replacing the switch, refer to the AS-04.08.039.



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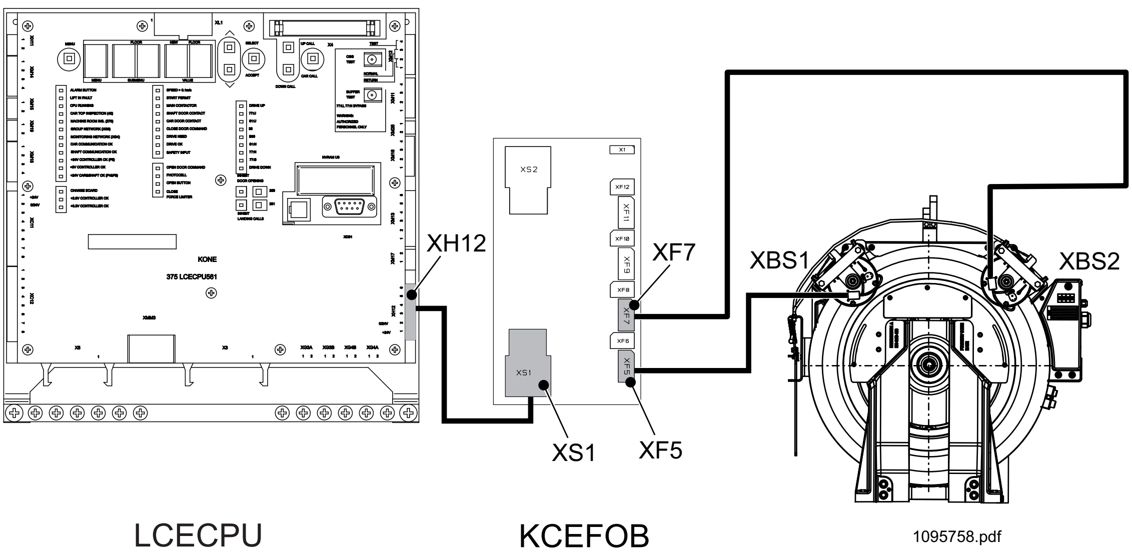
A.1 Wiring connection for LCE elevators



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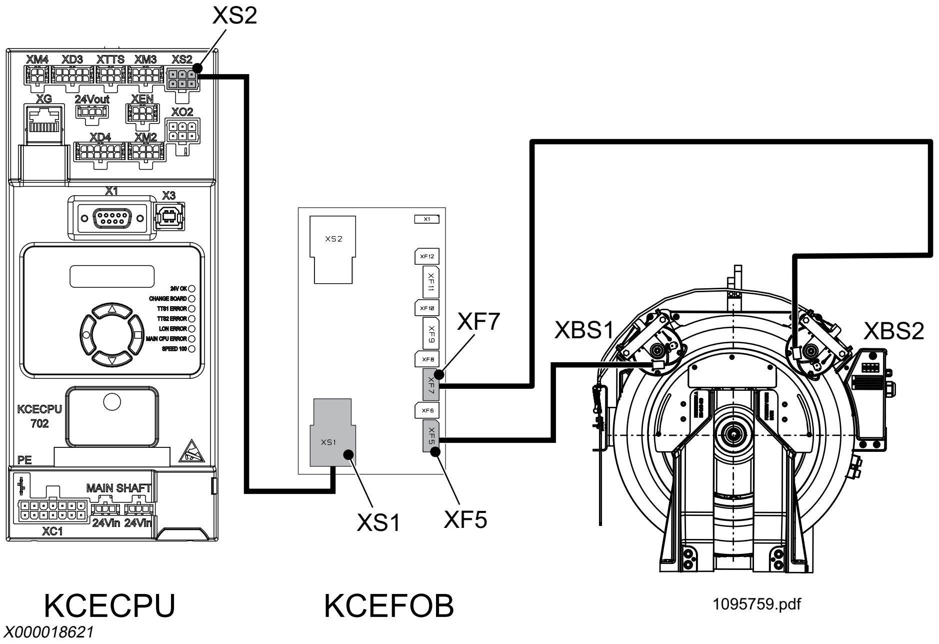
A.2 Wiring connection for KCE elevators



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APPENDIX B. SAFETY TESTS

Based on EN 81-20 requirements, the elevator must pass the following safety tests before it can be returned into service.

B.1 Safety tests with 0% load in car

B.1.1 One-sided electrical brake test (with 0% load)

The purpose of this test is to verify that one brake is capable of holding the car.

Before this test, elevator commissioning must be completed and the elevator must be driven in normal drive without failures. During this test, one brake lifts but the motor does not try to run.

Ensure that nobody can enter the elevator shaft or car during this test.

NOTE: LCE user interface opening tool KM878240G01 is needed to access the brake test parameters. The tool is available from KONE Global Spares Supply (GSS).

NOTE: The one-sided electrical brake test is applicable only for the following drive types:

– V3F16R, KDL16L, KDL16R, KDL16S

– KDL32, KDM

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B.1.1.1 One-sided electrical brake test with 0% load for LCE

1. Make a car call to the topmost floor from the control panel or controller.
2. Switch on the Recall Drive Feature (RDF).
3. Set the value of parameter 6_72 to 21 (brake 1 test).

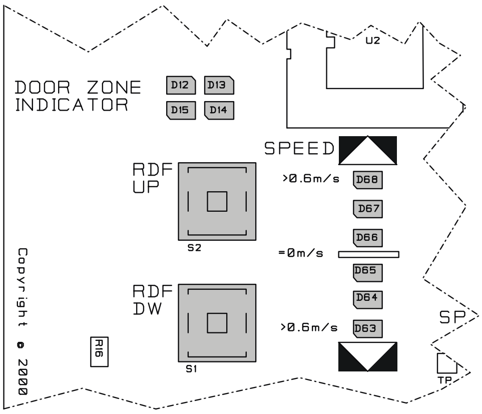
   NOTE: This activates the brake 1 test for one start only.

   NOTE: You may need the LCE user interface opening tool (KM878240G01) to access the parameter.

4. Push the RDF RUN and UP buttons.

   One brake opens for testing (brake 1 opened, brake 2 tested). The drive stops the test by itself after a maximum of 10 seconds. Monitor the SPEED LEDs at the same time.

   NOTE: The motor does not try to run. If the car moves, stop the test (release the RDF RUN and UP buttons)..



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5. Check the drive fault code from the error log.

   NOTE:

   – Drive fault code 126, subcode 6022 — test passed
   – Drive fault code 126 and subcode 2072 — test failed

   If the car moves or fault codes are displayed, check the following possible causes for brake 2 (the one that did not open during the test):

   – Brake shoe is mechanically stuck.
   – Incorrect counterweight / car balance
   – Dirt on the brake shoe
   – External lubricant leakage
   – Internal lubricant leakage
   – Wrong brake cable connection in the motor connection box

   Repeat the test. If the test fails again, replace the brakes or adjust the brakes.

6. Set the value of parameter 6_72 to 22 (brake 2 test).

   NOTE: This activates the brake 2 test for one start only.

7. Repeat the braking test with the other brake (brake 2 opened, brake 1 tested).

8. Check with drive code from the error log.

   NOTE:

   – Drive fault code 126, subcode 6021 — test passed

   – Drive fault code 126 and subcode 2071 — test failed

   If the car moves or fault codes are displayed, check the following possible causes for brake 1 (the one that did not open during the test):

   – Brake shoe is mechanically stuck.

   – Incorrect counterweight / car balance

   – Dirt on the brake shoe

   – External lubricant leakage

   – Internal lubricant leakage

   – Wrong brake cable connection in the motor connection box

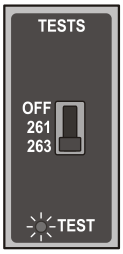
   Repeat the test. If the test fails again, replace the brakes or adjust the brakes.

WARNING: If any of the periodical tests fails, the elevator must not be put into normal use before it has been repaired and passed all the tests.

X0000188398 A.6

B.1.1.2 One-sided electrical brake test with 0% load for KCE

1. Set test switch to position 263. The test LED turns on.



X000102985

2. Switch on Recall Drive Feature (RDF).

3. Select 9_21 from KMI, and activate the test by setting the parameter value to 1.

   NOTE: Test is activated for one start only.

4. Drive the elevator car on RDF upwards for minimum 10 seconds.

   To drive RDF upwards, push and hold the RDF ‘RUN’ and ‘UP’ buttons together.

   One brake opens for testing (brake 1 opened, brake 2 tested).

   NOTE: The motor does not try to run.

   The drive stops the test by itself after 10 seconds (maximum).

   Monitor the SPEED LEDs on the KMI board at the same time.

5. Check the drive fault code from the error log.

   NOTE: Drive fault code d6022:

   – test passed

   Drive fault code d2083:

   – test failed

   If the car moves or drive fault codes are displayed, check the following possible causes for brake 2 (the brake that did not open during the test):

   – Brake centre nut is too tight (MX machines only)
   – Brake shoe is mechanically stuck.
   – Manual brake release wire is too tight.
   – Incorrect counterweight / car balance
   – Dirt on the brake shoe
   – External lubricant leakage
   – Internal lubricant leakage
   – Wrong brake cable connection in the motor connection box.

   Repeat the test. If the test fails again, replace the brakes.

6. Select 9_22 form KMI, and activate the test by setting the parameter value to 1.

   NOTE: Car must be still at the topmost floor.

7. Drive the elevator car on RDF upwards.

   To drive RDF upwards, push and hold the RDF ‘RUN’ and ‘UP’ buttons together.

   One brake opens for testing (brake 2 opened, brake 1 tested).

   NOTE: The motor does not try to run.

   The drive stops the test by itself after 10 seconds (maximum).

   Monitor the SPEED LEDs on the KMI board at the same time.

8. Check the drive fault code from the error log.

   NOTE: Drive fault code d6021:

   – test passed

   Drive fault code d2083:

   – test failed

   If the car moves or drive fault codes are displayed, see step 5 for the trouble shooting procedure.
9. Switch the elevator to normal drive.

X0000110335 A.15

B.1.2 Traction (stopping) test with empty car (with 0% load)

NOTE: Perform the test with empty car (with 0% load).

1. Give a car call to the lowest landing from the control panel.
2. Give a car call to the topmost landing from the control panel.
3. Wait for a short moment so that the elevator has reached the rated speed, then switch RDF on from the Recall Drive Unit.

   The elevator must stop.

   NOTE: The machine brakes must stop the car completely.

X0000189501 A.4

B.1.3 Traction (stalling) test with empty car (with 0% load)

NOTE: Perform the test with empty car (with 0% load).

1. Switch off the RDF.
2. Give a car call to the topmost landing from the control panel.
3. Switch on the RDF.
4. Drive the counterweight on the buffer.
5. When the counterweight has reached the buffer, continue driving the car up. Drive the car up with RDF for approximately three seconds.

   The car must not move.

   Check if the ropes are slipping on traction sheave while the traction sheave is rotating.
6. Drive the car back to the floor level by driving down with RDF.
7. Return the elevator to normal use.

X0000188424 A.5

B.2 Safety test with 100% load in car

B.2.1 Single brake test with rated speed and 100% rated load for LCE (EN 81-20 6.3.1 b)

NOTE: Before this procedure, all other brake tests must have been passed with 0% rated load.

NOTE: If the rated speed is greater than 2.5 m/s, set the elevator rated speed parameter to 2.5 m/s during the braking test.

1. Place 100% load in the car.
2. Inhibit the door opening and landing calls.

   Use LOP-CB switches 263 and 261.

   The following steps describe the testing of brake 1.
3. Drive the elevator car to the topmost floor.

   The test cannot be performed if the elevator car is not at topmost floor.
4. Switch on RDF (270).
5. Set Enable elevator test (6_72) parameter to 25 (brake 1 test).

   NOTE: This activates the brake 1 test for one start only.
6. Switch the elevator to normal drive.
7. Call the car to the bottom floor by using car calls.

   Drive engages one brake at middle of the elevator shaft.
8. Check the drive code from the error log.

   – Drive code 126 and subcode 6024_0_0:
     • test passed
   – Drive code 126 and subcode 2108_0_0:
     • test failed

   NOTE: Test fails also if:

   – any electric safety device is activated, even if drive code 126 and subcode 6024_0_0 are indicated.
   – any electric safety device is activated, even if no drive code is indicated.

9. If the fault code is displayed or any electric safety device is activated, check the possible fault causes for brake 1 and repeat the test:
   - Brake shoe is mechanically stuck.
   - Manual brake release wire is too tight.
   - Incorrect counterweight or car balance
   - Dirt on the brake shoe
   - External lubricant leakage
   - Internal lubricant leakage
   - Wrong brake cable connection in the motor connection box

   The following steps describe the testing of brake 2.
10. Drive the elevator car to the topmost floor.
11. Switch on the RDF (270).
12. Set Enable elevator test (6_72) parameter to 26 (brake 2 test).

    NOTE: This activates the brake 2 test for one start only.
13. Switch the elevator to normal drive.
14. Call the car to the bottom floor by using car calls.

    Drive engages one brake at middle of the elevator shaft.
15. Check the drive code from the error log.
    - Drive code 126 and subcode 6025_0_0:
      - test passed
    - Drive code 126 and subcode 2109_0_0:
      - test failed

    NOTE: Test fails also if:
    - any electric safety device is activated, even if drive code 126 and subcode 6025_0_0 are indicated.
    - any electric safety device is activated, even if no drive code is indicated.

16. If the fault code is displayed or any electric safety device is activated, check the possible fault causes for brake 2 and repeat the test:
    - Brake shoe is mechanically stuck.
    - Incorrect counterweight or car balance
    - Dirt on the brake shoe
    - External lubricant leakage
    - Internal lubricant leakage
    - Wrong brake cable connection in the motor connection box
    - Incorrect brake adjustment

    NOTE: If the test fails again, replace the brake.

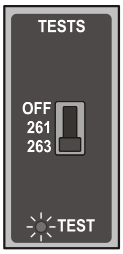
X0000110307 B.3

B.2.2 Single brake test with rated speed and 100% rated load for KCE (EN 81-20 6.3.1 b)

NOTE: Before this procedure, all other brake tests must have been passed with 0% rated load.

NOTE: If the rated speed is greater than 2.5 m/s, set the elevator rated speed parameter to 2.5 m/s during the braking test.

1. Place 100% load in the car.
2. Drive the elevator to topmost floor.
3. Set the test switch to position 263.



X000102985

4. Switch the RDF on.
5. Select 9_25 from KMI, and set the parameter value to 1 to activate the test.
6. Switch the RDF off.
7. Give a car call to bottom floor using KMI.

   NOTE: Call needs to be given within 30s after activating the test from KMI.

   Elevator starts to drive. In the middle of the elevator shaft, one brake is engaged.

8. Check the drive fault code from the error log.

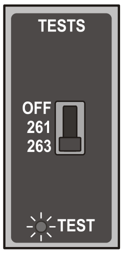
   Drive fault code d6024:

   – test passed

   Drive fault code d2108:

   – test failed

9. Switch elevator to normal drive.
10. Drive the elevator to topmost floor.
11. Set test switch to position 263.

    X000102985

    The test LED turns ON.
12. Switch the RDF on.
13. Select 9_26 from KMI, and set the parameter value to 1 to activate the test.
14. Switch the RDF off.
15. Give a car call to bottom floor using KMI.

    Elevator starts to drive. In the middle of the elevator shaft, one brake is engaged.

    NOTE: Call needs to be given within 30s after activating the test from KMI.

16. Check the drive fault code from the error log.

    Drive fault code d6025:

    – test passed

    Drive fault code d2109:

    – test failed

17. Switch the elevator to normal drive.

X0000110336 B.2

B.3 Safety test with 125 % load in car

B.3.1 Brake test with rated speed and 125% rated load for KDL16 with LCE (EN 81-20 6.3.1 b)

NOTE: Before this procedure all other brake tests must pass 0% rated load and 100% rated load.

NOTE: For elevators where the nominal speed is over 2.5 m/s, set the elevator nominal speed parameter to 2.5 m/s during the braking test with 125% load.

1. Add 125% load weight inside the car.
2. Inhibit door opening and landing calls.

   Use LOP-CB switches 263 and 261.
3. Switch elevator to RDF mode.
4. Select **Enable elevator test (6–72)** to **24 (enhanced traction test)**.
5. Switch elevator to normal mode.
6. Give a car call to the bottom floor using LOP — CB

   WARNING: Do not give up a car call.

   ---

   1. Elevator drives downwards and makes a brake stop in the middle of the elevator shaft.
   2. After stop, the elevator makes a correction drive.
   3. Elevator drives to floor.

      If the fails, replace the brakes or adjust the brakes.

X0000190570 A.6

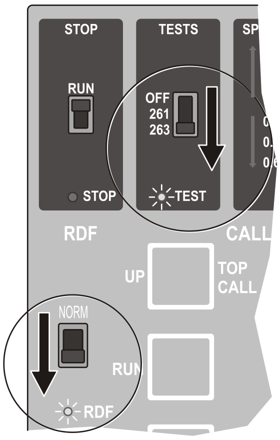
B.3.2 Braking test with rated speed and 125% rated load for KDL16 and KDM with KCE (EN 81-20 6.3.1 b)

NOTE: Before this procedure, all other brake tests must have been passed with 0% rated load and 100% rated load.

NOTE: For elevators where the nominal speed is over 2.5 m/s, set the elevator nominal speed parameter to 2.5 m/s during the braking test with 125% load.

1. Add weights inside the car (125 % load).

2. Set test switch to position 263.



X000103031

The test LED turns ON.

3. Switch the elevator to RDF.
4. Select 9_24 from KMI.

   Activate the 125% traction test by setting the parameter value to 1.
5. Switch elevator to normal mode.
6. Press boost in KMI.
7. Give a car call to bottom floor using KMI.

WARNING: Do not give a call call up.

---

1. Elevator drives downwards and makes a brake stop in the middle of the shaft.
2. After stop the elevator makes correction drive.
3. Elevator drives to floor.

   If the fails, replace the brakes or adjust the brakes.

X0000184750 A.9

B.3.3 Braking test with rated speed and 125% rated load for KDM with LCE (EN 81-20 6.3.1 b)

NOTE: Before this procedure, all other brake tests must have been passed with 0% rated load and 100% rated load.

NOTE: For elevators where the nominal speed is over 2.5 m/s, set the elevator nominal speed parameter to 2.5 m/s during the braking test with 125% load.

1. Add weight inside the car (125% load).

2. Inhibit landing calls and door opening.
3. Drive the elevator car to the topmost floor.
4. Switch elevator to RDF mode (270 on).
5. Set elevator nominal speed parameter (6–3) to 2.5 m/s.

   NOTE: Set the value to elevator nominal speed if it is slower than 2.5 m/s.

6. Select ReatTimeDisplay monitor selection (6–75) to 1 (Elevator speed).
7. Switch elevator to normal drive (270 off).
8. Give a car call down using LCECPU.
9. When the elevator has reached the rated speed, push the stop switch.

   If there is no stop switch, stop the car by switching the elevator to RDF.

   NOTE: The elevator car must stop.

   If the fails, replace the brakes or adjust the brakes.

X0000188390 A.8

X0000189147 A.4

APPENDIX C. DE-ENERGIZING

C.1 De-energize KDL16R drive

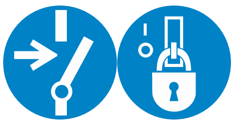
Multimeter is needed for the measurements. Use CAT III-compliant multimeter. Check the operation of the multimeter before and after each measurement.

WARNING: Risk of electric shock

WARNING: Inverter drives usually remain energized for about 5 minutes after the power is disconnected. Do not work on the drive, hoisting machine or braking resistors until you have checked that this energy has been discharged. Set the test equipment to 1000 VDC range. Check the test equipment before and after the test to ensure that it functions correctly.

---

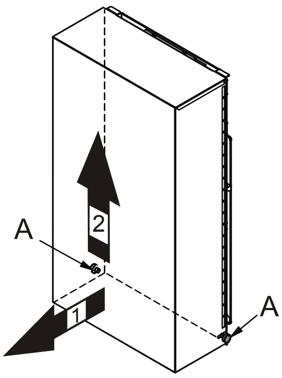
1. Disconnect and prevent reconnection of the possible emergency power supplies.
2. Switch off the main switch.

   Lock and tag.

   NOTE: If the elevator has a remote main switch, it must be switched off also.

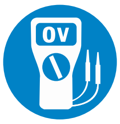
3. Wait for 5 minutes.
4. Remove the cover.

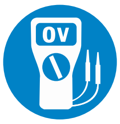
   X000017085

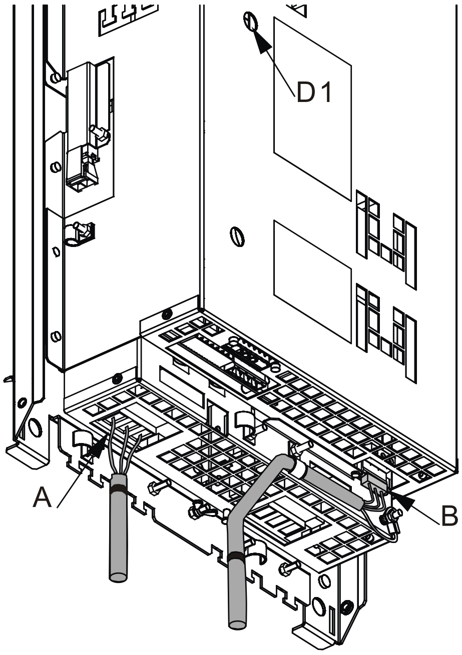
5. Set the multimeter to 1000 VAC range.

6. Measure that there is no voltage (AC) in power supply terminals (T1, T2, T3).
    1. Measure between each phase and protective earth: T1-PE, T2-PE, T3-PE.
    2. Measure between phases T1-T2, T1-T3, T2-T3.



7. Set the multimeter to 1000 VDC range.
8. Measure that there is less than 20 VDC voltage in the intermediate circuit from braking resistor terminal XBRE2 (B).
    1. Measure from the connector XBRE2 (B), between terminals: 1 and PE
    2. Measure from the connector XBRE2 (B), between terminals: 3 and PE





X000015353

9. Install the drive protective cover.

X0000191405 B.2

Related information

– *Remove old machine (25)*

C.2 De-energize KDL16S drive

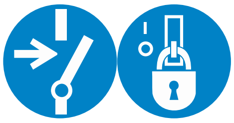
Multimeter is needed for the measurements. Use CAT III-compliant multimeter. Check the operation of the multimeter before and after each measurement.

WARNING: Risk of electric shock

WARNING: Inverter drives usually remain energized for about 5 minutes after the power is disconnected. Do not work on the drive, hoisting machine or braking resistors until you have checked that this energy has been discharged. Set the test equipment to 1000 VDC range. Check the test equipment before and after the test to ensure that it functions correctly.

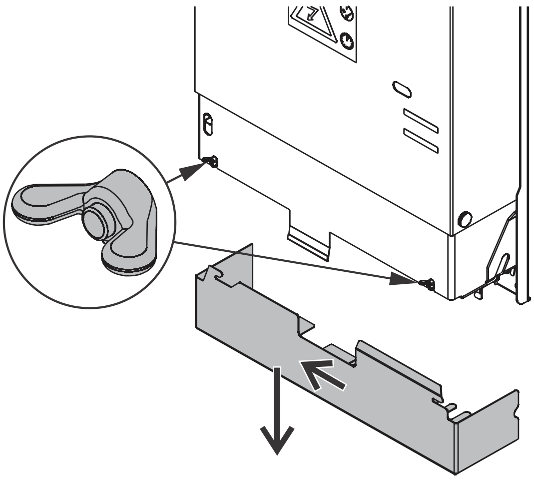
---

1. Disconnect and prevent reconnection of the possible emergency power supplies.
2. Switch off the main switch.

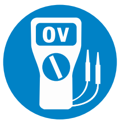
   Lock and tag.

   NOTE: If the elevator has a remote main switch, it must be switched off also.
3. Wait for 5 minutes.
4. Remove the drive protective cover.

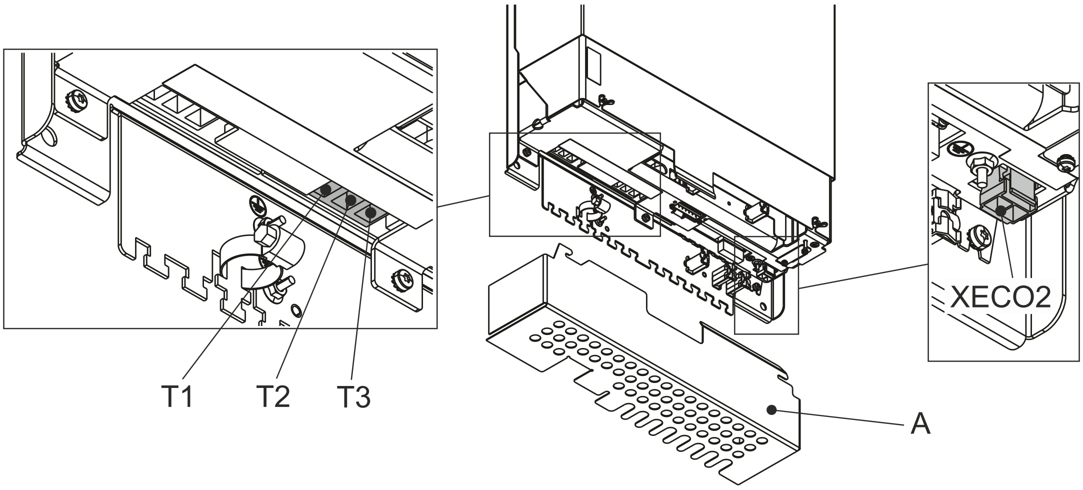
   

   X0000111015
5. Set the multimeter to 1000 VAC range.

6. Measure that there is no voltage (AC) in power supply terminals (T1, T2, T3).
    1. Measure between each phase and protective earth: T1-PE, T2-PE, T3-PE.
    2. Measure between phases: T1-T2, T1-T3, T2-T3.



7. Set the multimeter to 1000 VDC range.
8. Measure that there is less than 20 VDC voltage in the intermediate circuit from ECB-1 output.
    1. Measure from the connector XECO2, between terminals: 1 and PE
    2. Measure from the connector XECO2, between terminals: 2 and PE
    3. Measure from the connector XECO2, between terminals: 1 and 2



X0000228719

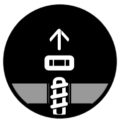
9. Install the protective wire cover (A).

X0000200387 E.2

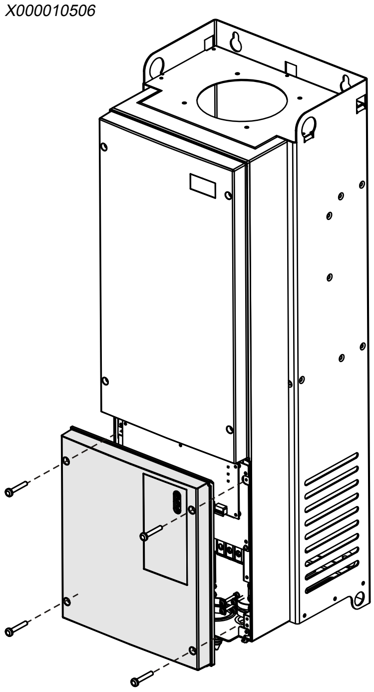
C.3 De-energize the system when KDM drive

1. Switch off main power.
2. Wait for 5 minutes.

3. Remove the fixing screws (4 pcs.) and cover of the drive module.

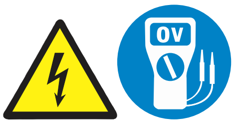


X000010506



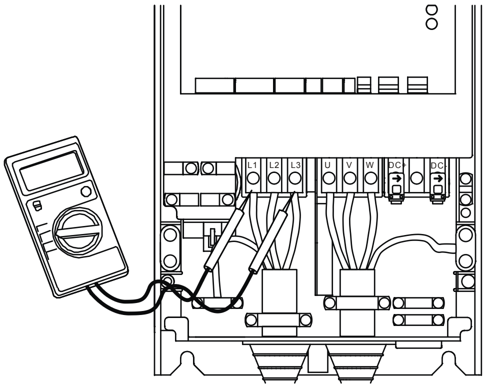
X000015931

4. Check that there is no voltage (AC) in the drive module by performing the following actions:

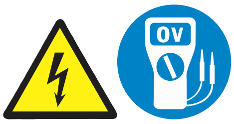
   1. Set the multimeter to 1000-VAC range.
   2. Measure with multimeter from power supply terminals, between:
      – Main phases L1-L2, L1-L3, L2-L3
      – Main phases against PE

   NOTE: Verify multimeter functionality.

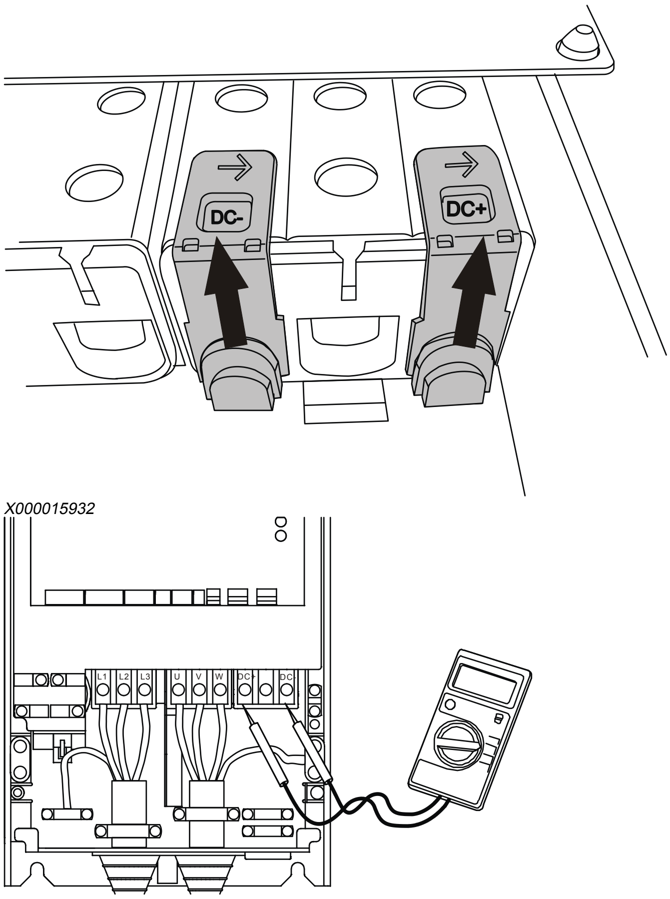
   

   X000020386

5. Check that there is no voltage (DC) in the intermediate circuit by performing the following actions:



   1. Set the multimeter to 1000-VDC range.
   2. Remove the protective covers from intermediate circuit terminals.
   3. Measure DC voltage between DC+ and DC- terminals.



6. Replace the protective covers to the intermediate circuit terminals DC+ and DC-.

X0000188154 A.6